	GLOBAL ECONOMICS **Grade: 10 - Learning Evidence 3** **DECISION-MAKING UNDER UNCERTAINTY** **Teacher: Nicolás López Cuéllar**	**1st TERM** **2026–2027**

Learning objective: Compare decision criteria and their limitations, and construct and interpret decision trees for staged filtering under uncertainty.	**Assessment criteria:** **C8**: Compares the Maximax, Maximin, Minimax Regret, and EMV/Bayes decision criteria by contrasting their information requirements, risk orientations, recommended alternatives, and limitations. **C9**: Constructs and interprets decision trees for staged filtering rules, applying decision criteria in the specified order to identify the final alternative. **C10**: Evaluates economic and financial decision-making scenarios by selecting and justifying the most appropriate decision criterion according to the information available, risk attitude, and decision context.

Name: _____ **Grade:** _____ **Date:** _____

1. Criteria assessment

This exam assesses C8 and C9, each worth 6 points. A **minimum of 4 points per criterion** is required to pass. Show all relevant calculations and justify your answers. Partial credit may be awarded for correct procedures despite arithmetic errors; answers without demonstrated procedures receive no credit. A 1-point penalty per criterion applies if the student's name is missing or the assessment is submitted late. C10 is assessed separately.

Question	**C8**	**C9**
1	3	–
2	3	–
3	–	2
4	–	2
5	–	2
Total	**6**	**6**

2. 1. Repair Workshop Plans [C8-3]

A bicycle repair workshop selects its capacity under uncertain demand. Profits are in thousand USD.

Alternative	High	Normal	Low
$P(S_j)$	0.2	0.6	0.2
Large (A_1)	120	25	-20
Flexible (A_2)	85	65	-10
Basic (A_3)	50	45	35
Partner (A_4)	70	75	20

Decision statistics (Regret = maximum regret):

Alt.	Max	Min	Regret	EMV
A_1	120	-20	55	35
A_2	85	-10	45	54
A_3	50	35	70	44
A_4	75	20	50	63

(a) Identify each criterion's recommendation and deciding statistic value. **[1]**

(b) A subsidy s (integer, $0 \leq s \leq 50$) increases the low-demand payoffs of A_1 and A_2 by s. An alternative is eligible only if all its payoffs are nonnegative. Determine each alternative's eligibility based on the value of s and find the smallest s making A_2 the sole Maximin choice. **[2]**

3. 2. Community Market Stall [C8-3]

An organizer selects a market stall under uncertain attendance. Profits are in hundred USD.

Alternative	High	Low
$P(S_j)$	0.4	0.6
Large (C_1)	80	10
Compact (C_2)	48	32
Standard (C_3)	70	25
Premium (C_4)	90	-5

Decision statistics (Regret = maximum regret):

Alternative	Max	Min	Regret	EMV
C_1	80	10	22	38
C_2	48	32	42	38.4
C_3	70	25	20	43
C_4	90	-5	37	33

(a) Identify each criterion's recommendation and deciding statistic value. **[1]**

(b) A sponsor replaces C_2's high-attendance payoff of 48 with g (integer, $48 \leq g \leq 80$). An alternative is eligible only if all its payoffs are at least 20. Determine each alternative's eligibility based on g and find the smallest g making C_2 the sole Minimax Regret choice among eligible alternatives. **[2]**

4. 3. Book Fair Stock [C9-2]

A bookseller selects a stock plan D_1–D_6. Net profits are in hundred USD. The decision statistics are summarized below:

Alternative	Max	Min	Regret	EMV
D_1	100	0	40	65
D_2	90	10	30	65
D_3	80	20	20	65
D_4	70	30	30	59
D_5	60	40	40	53
D_6	50	30	50	43

Apply these stages in order:

1. EMV/Bayes: Best 3.
2. Maximax: Best 2.
3. Maximin: Final selection.

Illustrate each filtering stage, showing survivors and eliminations.

5. 4. Refill Shop Expansion [C9-2]

A refill shop selects an expansion plan F_1–F_6. Net profits are in thousand USD. The decision statistics are summarized below:

Alternative	Max	Min	Regret	EMV
F_1	120	0	45	61
F_2	105	15	30	64.5
F_3	95	25	25	68.5
F_4	85	35	35	70
F_5	75	45	45	64
F_6	65	40	55	55

Apply these stages in order:

1. Minimax Regret: Best 5.
2. EMV/Bayes: Best 4.
3. Maximax: Best 2.
4. Maximin: Final selection.

Illustrate each filtering stage, showing survivors and eliminations.

6. 5. School Supply Orders [C9-2]

A school supplier selects an order plan H_1–H_6. Net profits are in hundred USD. The decision statistics are summarized below:

Alternative	Max	Min	Regret	EMV
H_1	120	0	45	61
H_2	105	15	30	64.5
H_3	95	25	25	68.5
H_4	85	35	35	70
H_5	75	45	45	64
H_6	65	40	55	55

Apply these stages in order:

1. Maximin: Best 4.
2. Minimax Regret: Best 3.
3. Maximax: Best 2.
4. EMV/Bayes: Final selection.

Illustrate each filtering stage, showing survivors and eliminations.

7. C10: Separate activity reminder (unscored)

Complete the individual **Interactive Decision Webpage: C10 Practice** activity and submit the deployed webpage URL. See the separate activity for instructions.